

L1 - Heaps

Special class

~~1st~~

Heap $\rightarrow$?



D.S



[C.B.T]

?

Binary tree

Parent

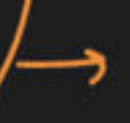
Max Min

+

[Heap property satisfy]

?

C.B.T



Complete Binary tree

① ^{may or may not be}
except last level

$\rightarrow$ all levels are completely filled

② all the nodes in the last level

filled from left $\rightarrow$ right [continuous manner]

Heap $\rightarrow$ DS $\rightarrow$ (BT + Heap property)
Man/Min

①
②

Heap $\rightarrow$ visualize $\rightarrow$ Tree
Impl $\rightarrow$ Array

0 - band $\rightarrow$ i
 $2i+1$ $2i+2$

1 - band $\rightarrow$ i
 $2i$ $2i+1$

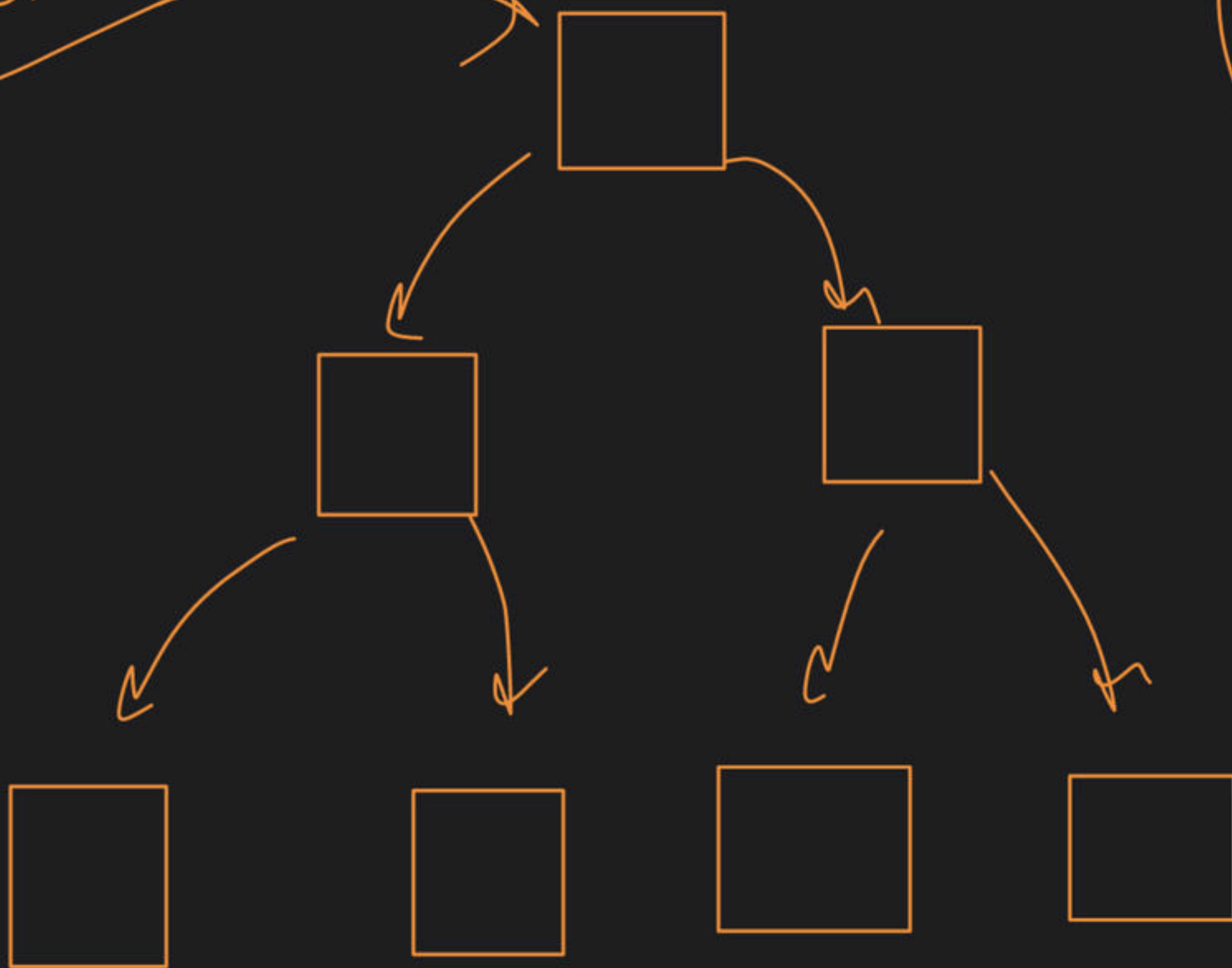
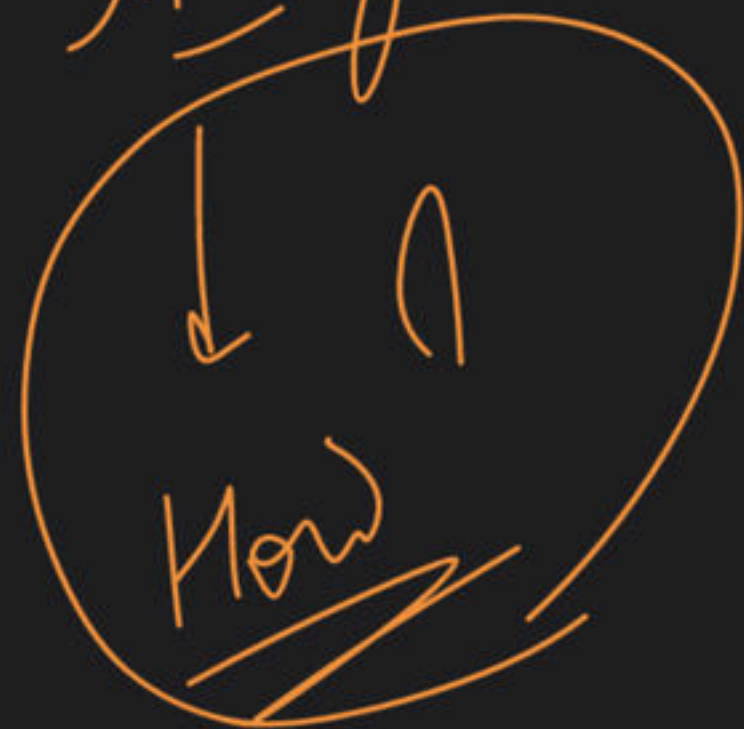
parent
child
child
 $\frac{\text{child}}{2}$

Heap

simulation

Impl

Array



Max-Heap

$P > \text{child}$

Min-Heap

$P < \text{child}$

Heap → Array

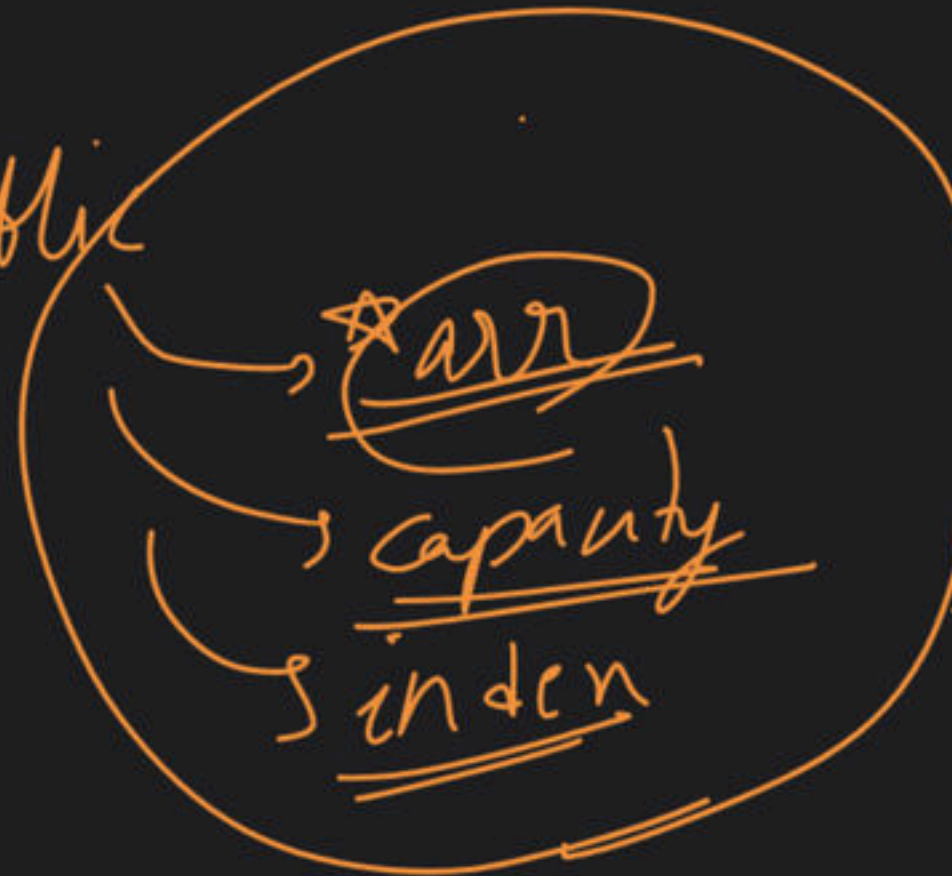
→ insert

→ delete

Class Heap

{

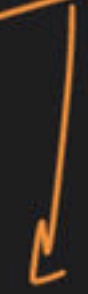
public



→ ✓

}

Heapsort

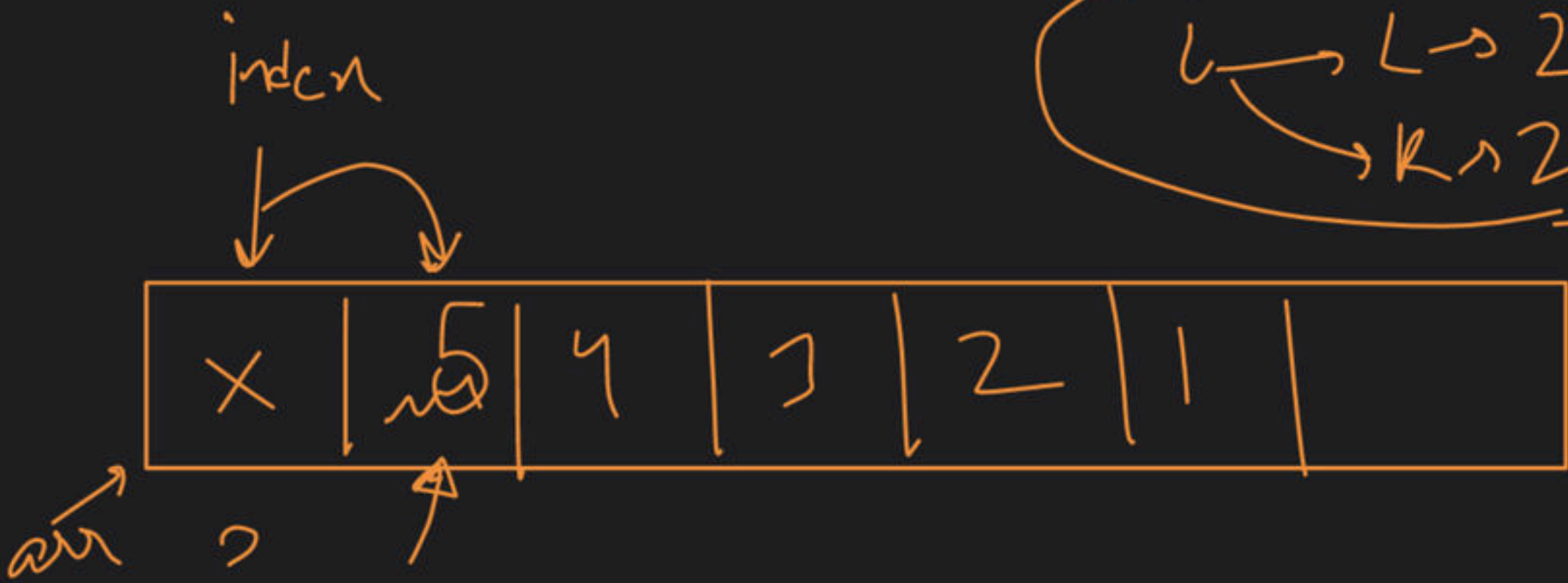


Heap

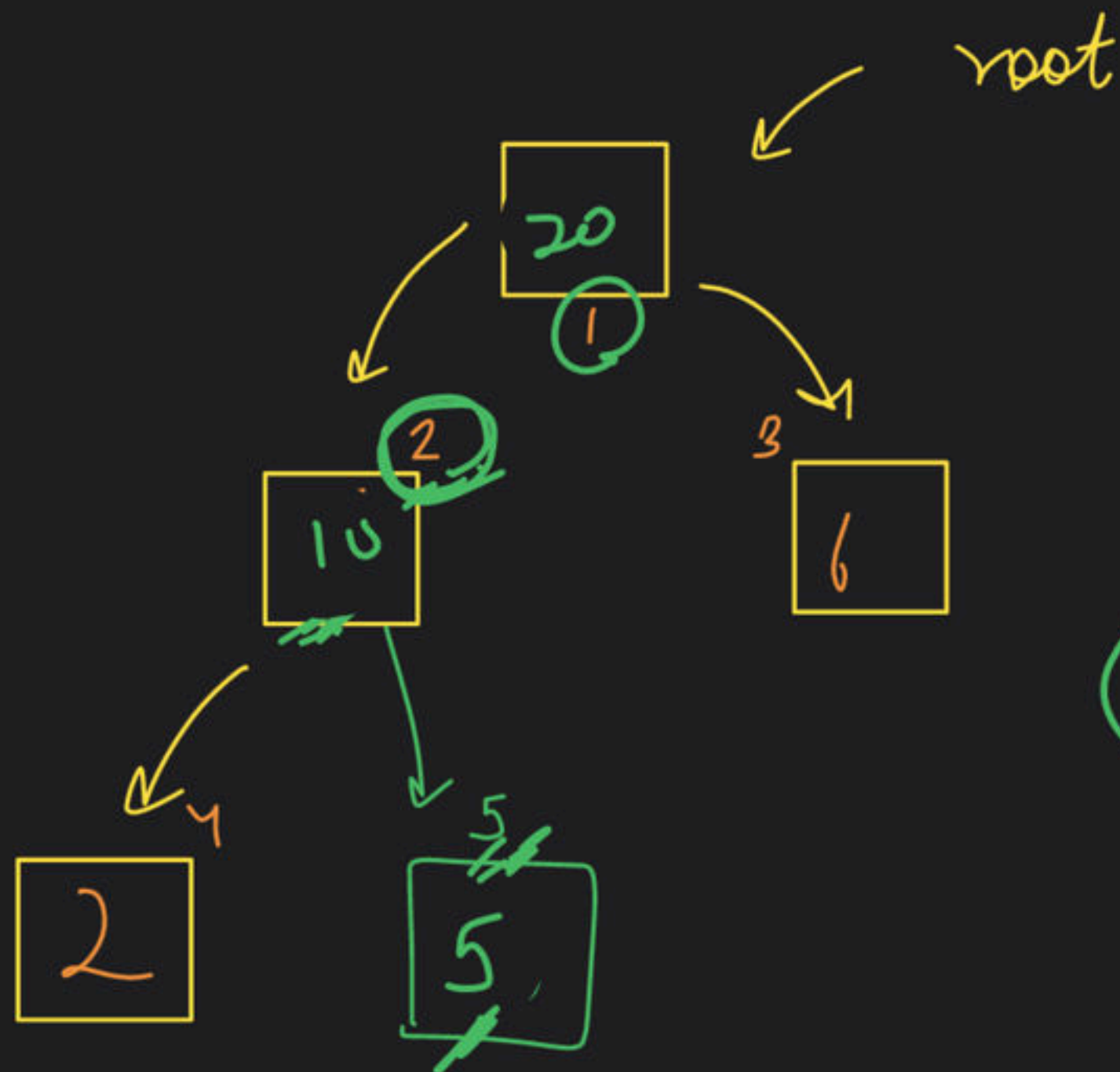
insert

- ① Overflow
- ② insert logic
- ③ element

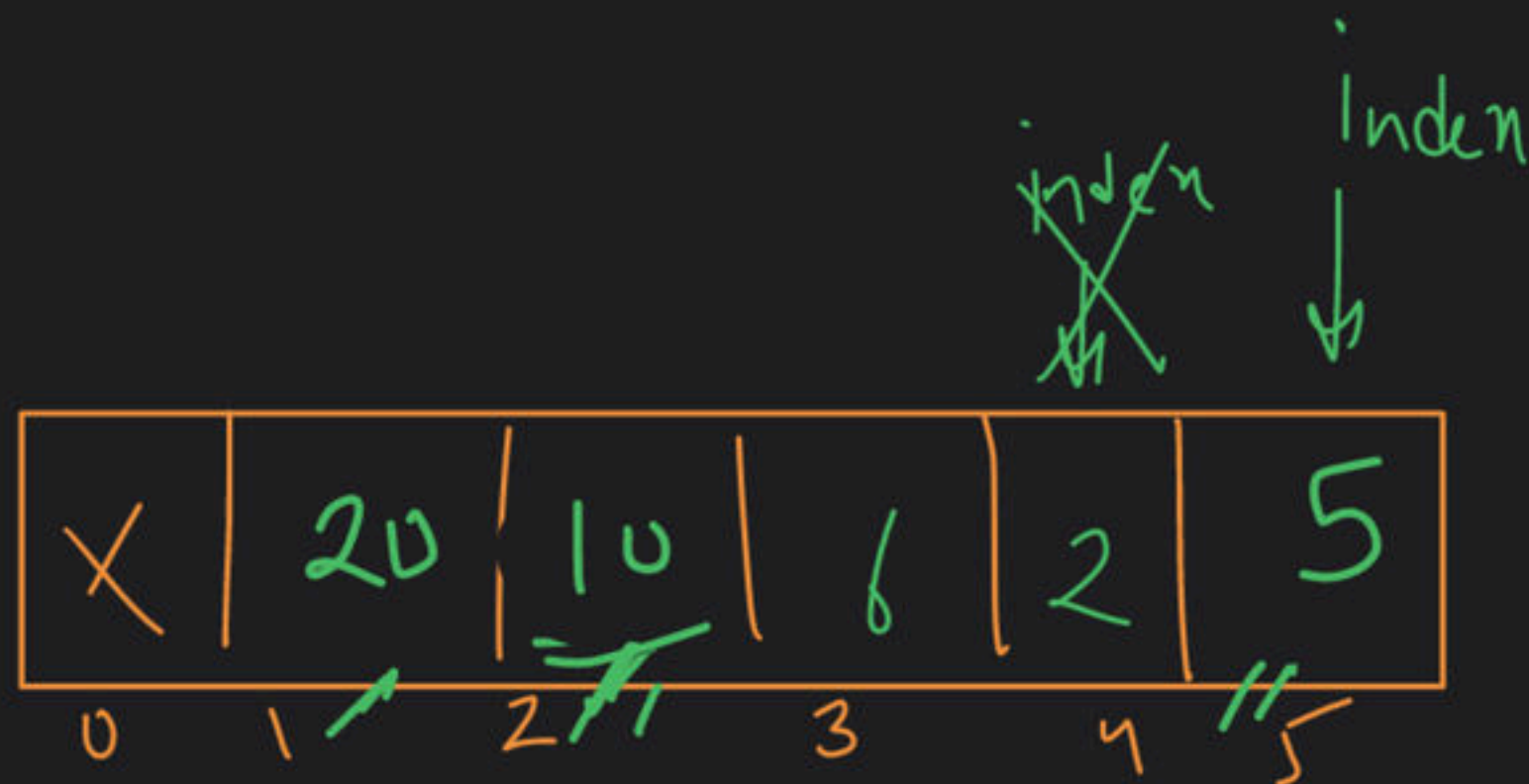
→ $index++$ → $arr[index] = val$
correct
right position
why



$i \rightarrow L \rightarrow 2i$
 $\quad \rightarrow R \rightarrow 2i+1$



$i = 2$
 $p = 1$
 $i = p$
 $i = 1$

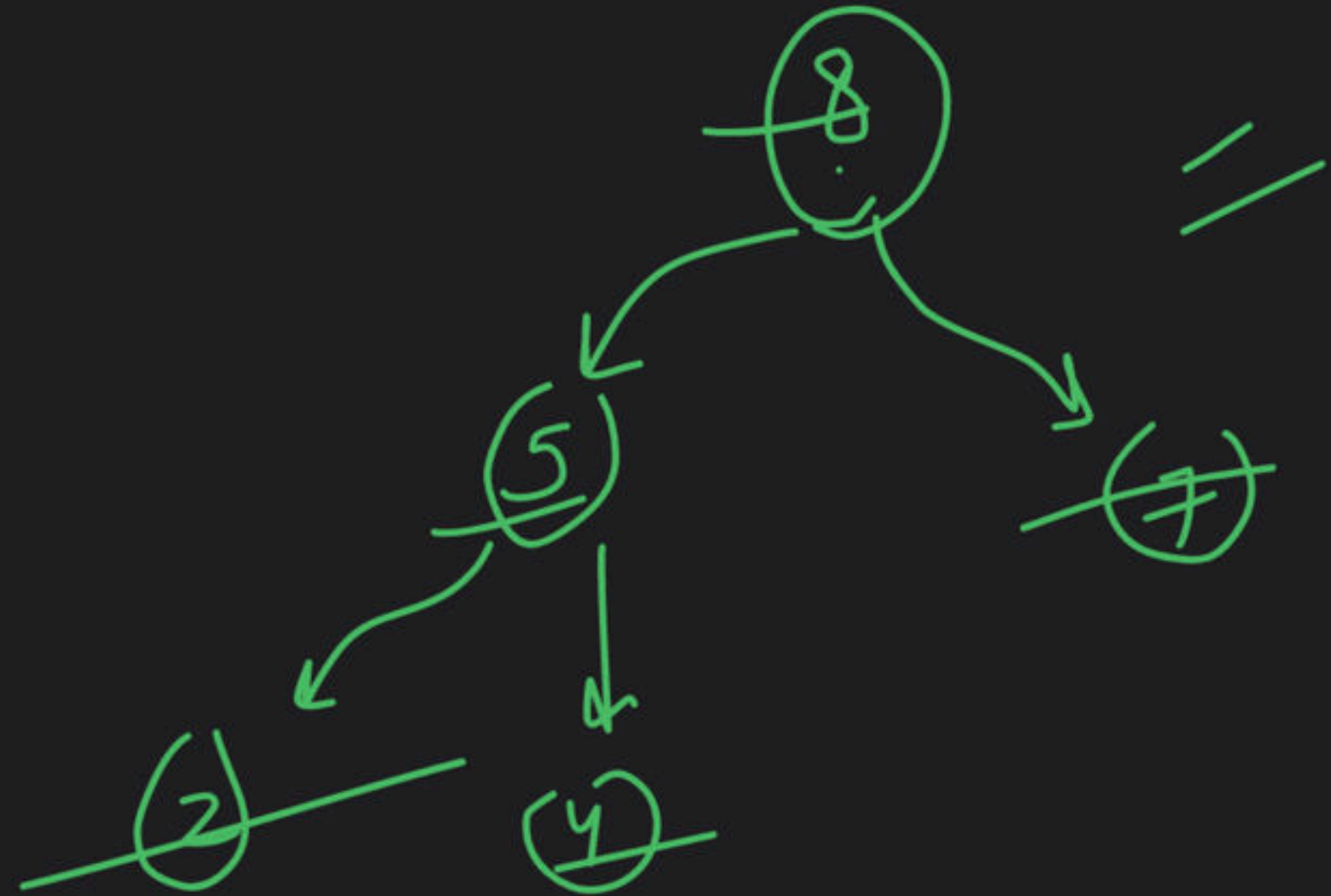


while ($i > 1$)
 {
 parentIndex = $i/2$;
 if ($arr[parentIndex] < arr[i]$)
 {
 swap ($arr[parent]$, $arr[i]$)
 $i = parentIndex$;
 }
 else
 {
 break;
 }
 }

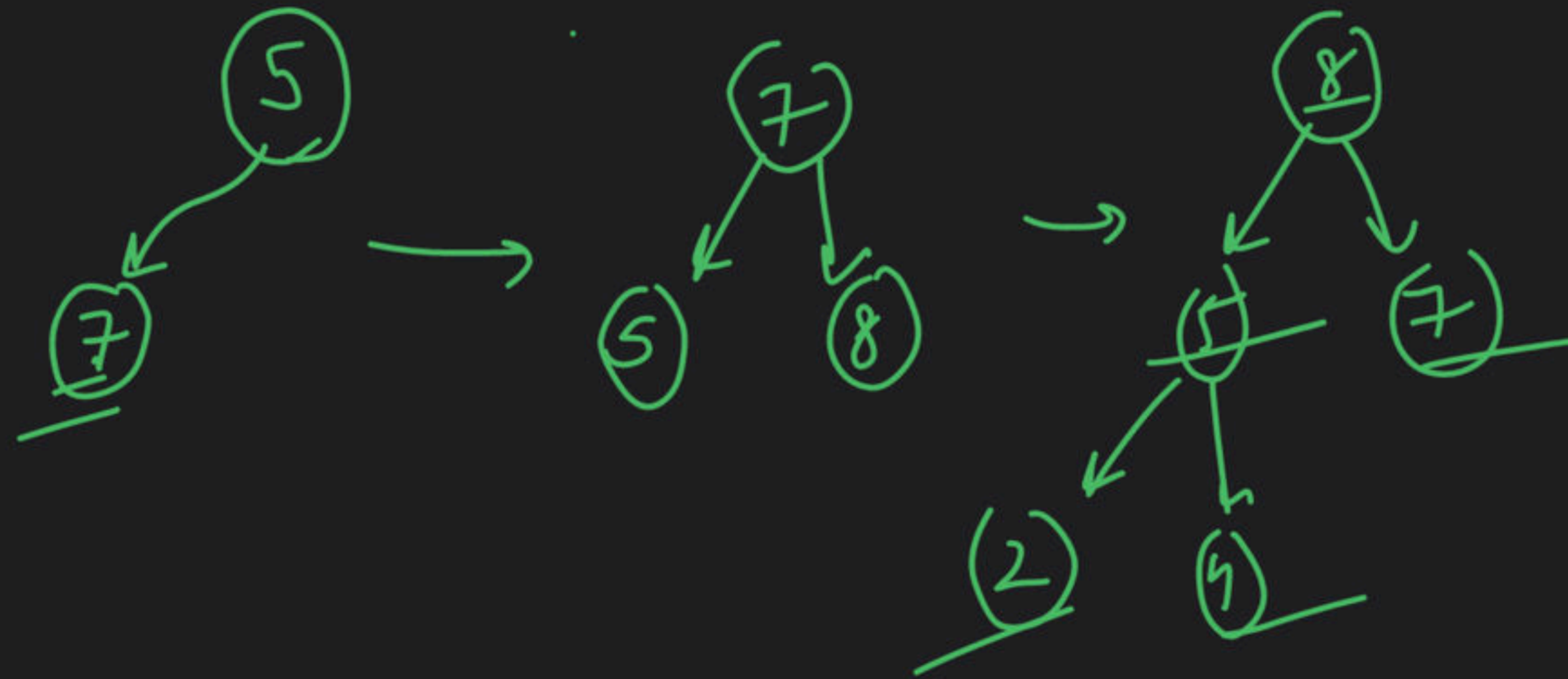
Annotations:

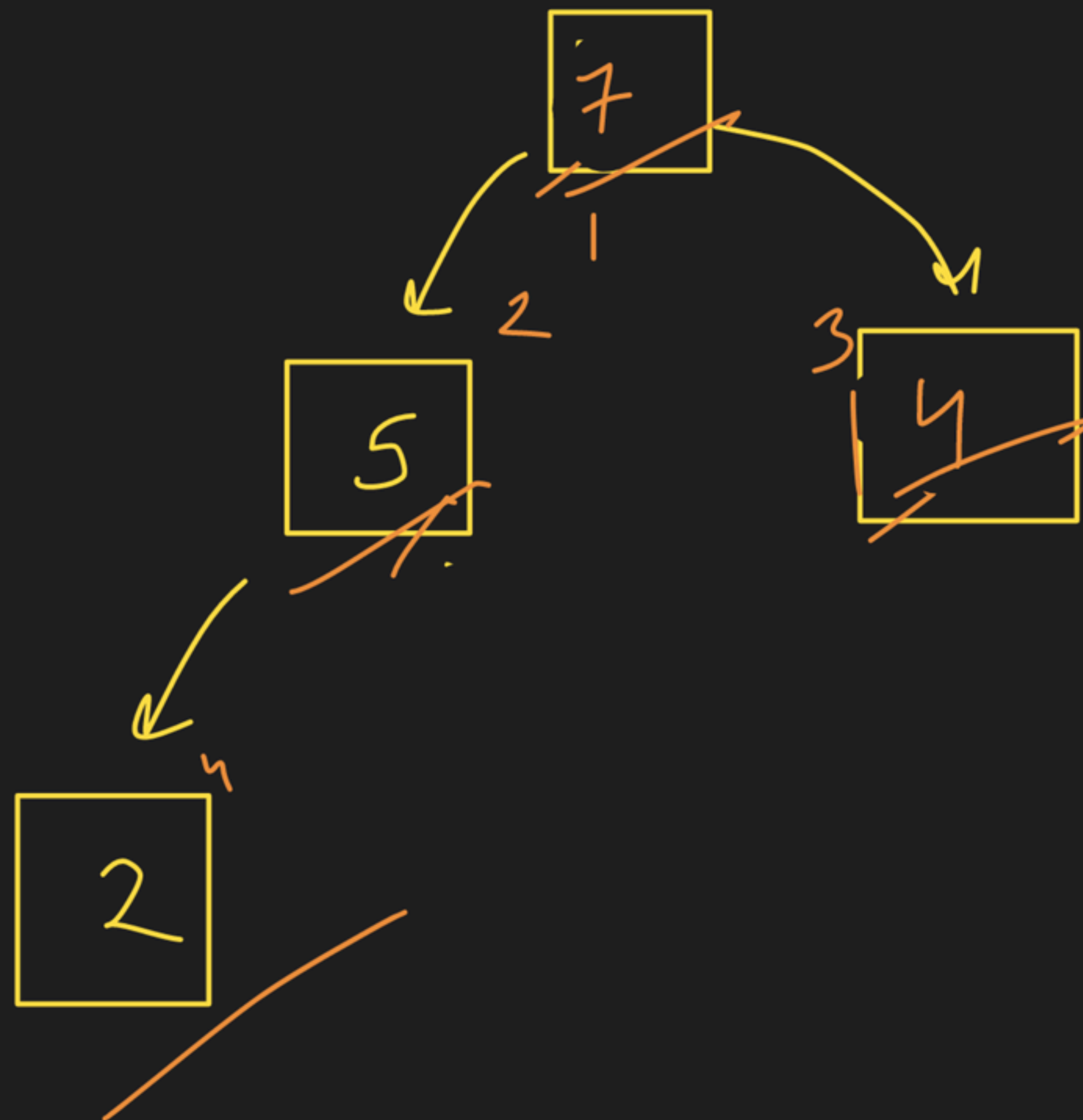
- "correct pos" points to the while loop condition.
- "why" points to the assignment $i = parentIndex$.
- "Bhot" points to the break statement.

x	8	5	7	2	4
0	1	2	3	4	5



5 7, 8, 2, 1





array $\rightarrow$ ilp



Heap

~~for (n/2 $\rightarrow$ 1)~~

heapify



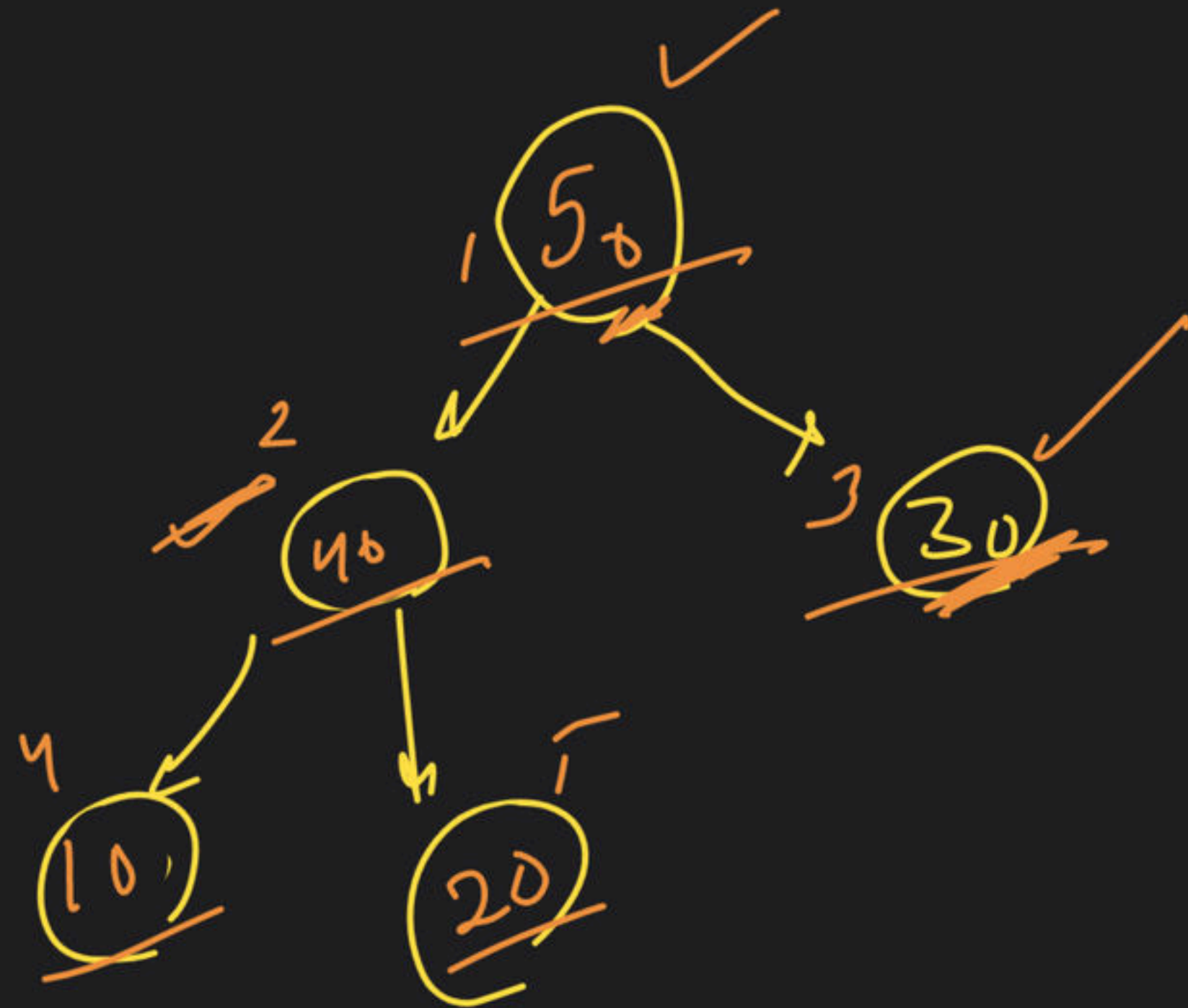
heapify

X	10	20	30	40	50	60
0	1	2	3	4	5	6

$n = 6$

$n/2 = 3$

for $n/2 \rightarrow 1$

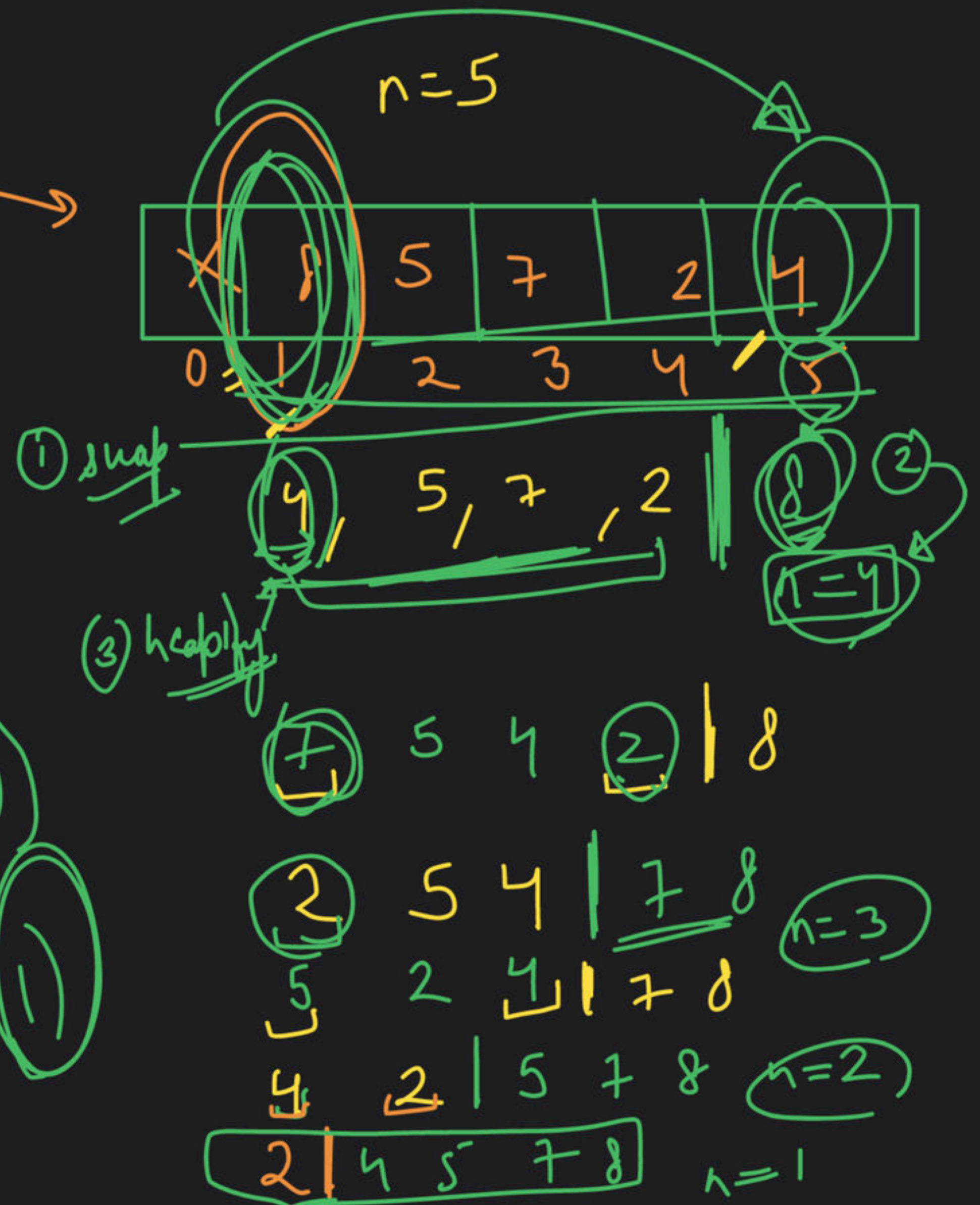
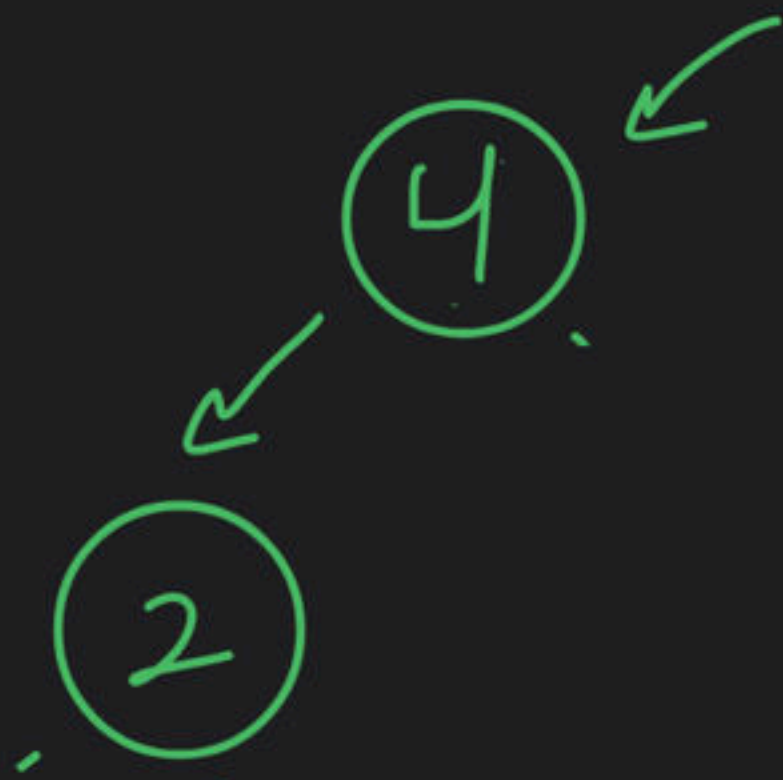


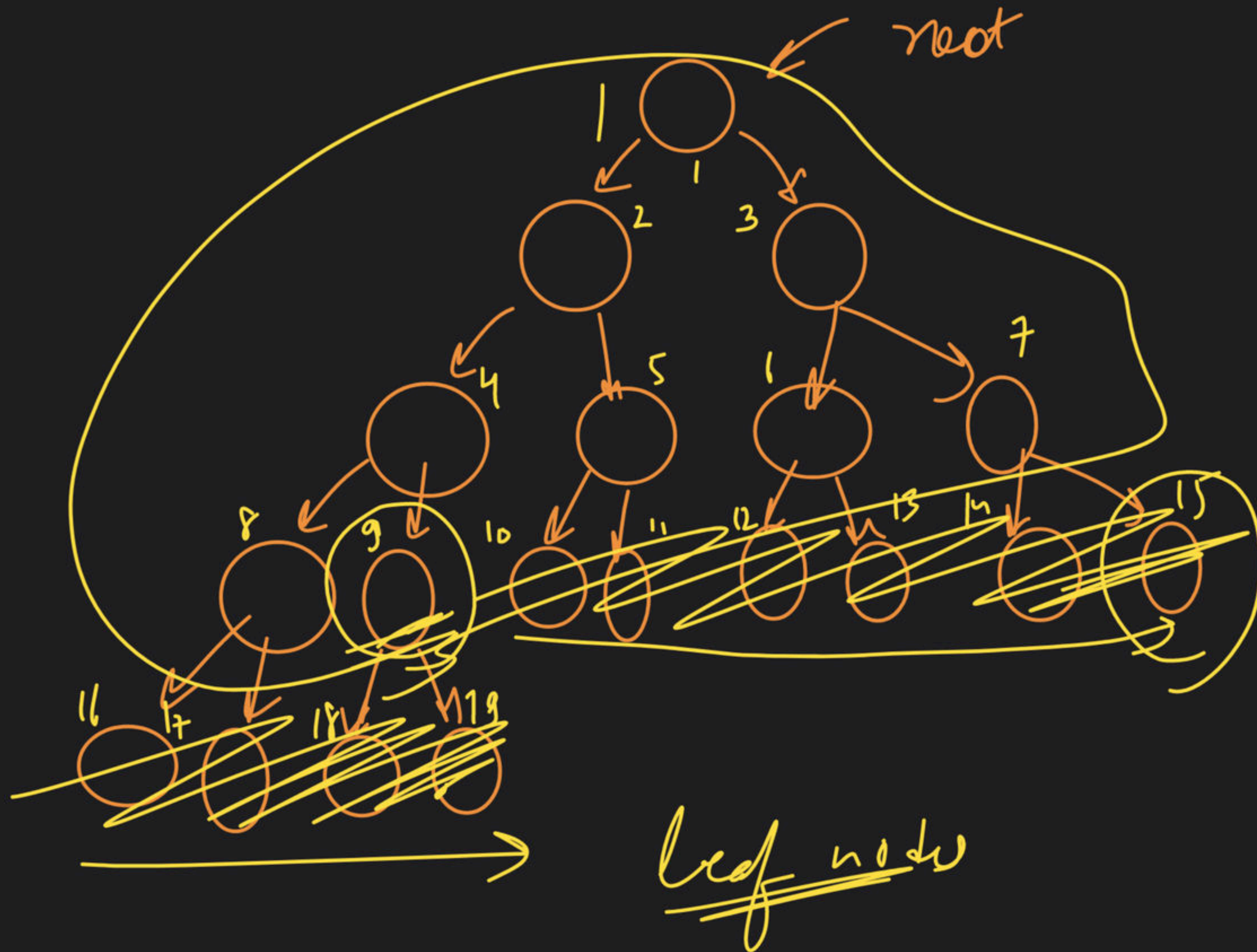
-1	10	20	30	40	50
----	----	----	----	----	----

Heap sort

- (1) Build heap
- (2) Sorting algo

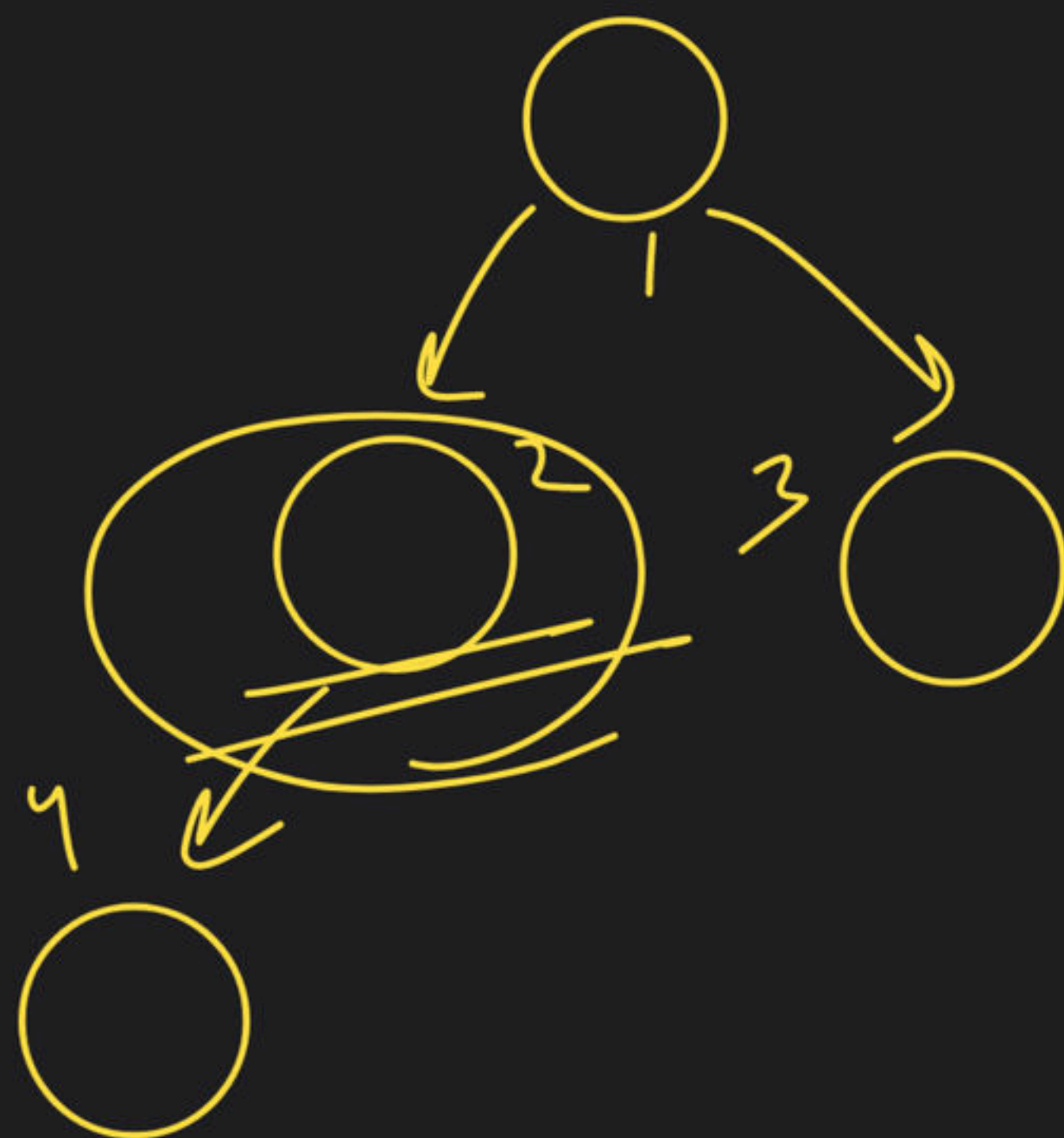
$\text{while } (n > 1)$
 $\{$
 $\text{swap } (\text{arr}[1], \text{arr}[n])$
 $n--$
 $\text{heapify}(\text{arr}, n, 1)$
 $\}$





$$n = 19$$
$$n/2 \rightarrow 9$$

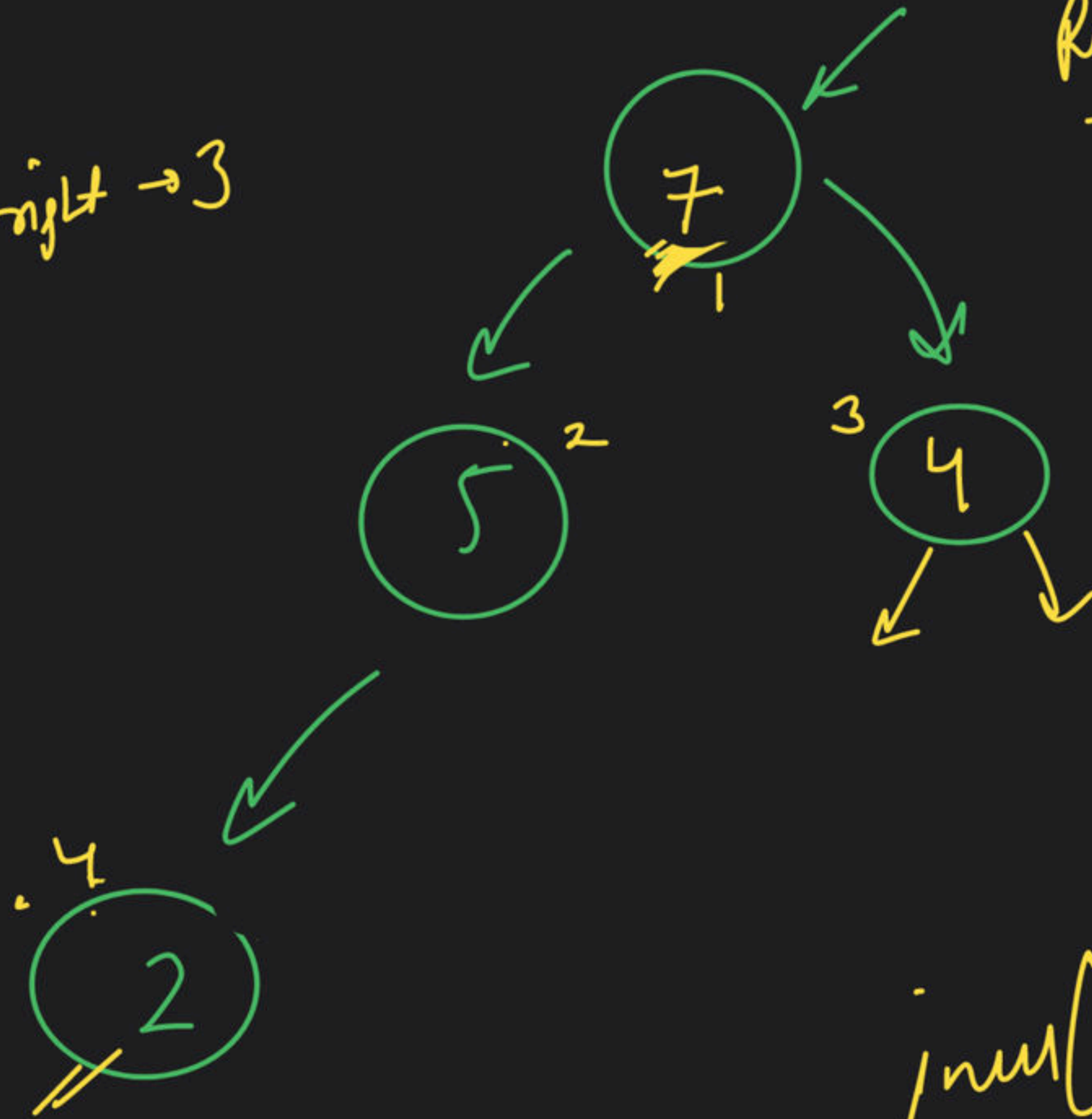
$$1 \rightarrow 9$$
$$1 \rightarrow 19$$



$$n = 4$$

$$n/2 \rightarrow 2$$

$p \rightarrow 1$
 $\text{left} \rightarrow 2$ $\text{right} \rightarrow 3$



Rukna Kab h $\rightarrow$ valid position
 $\rightarrow$ child α

pq.pop()

① swap [arr[i], arr[inden]]

② inden --

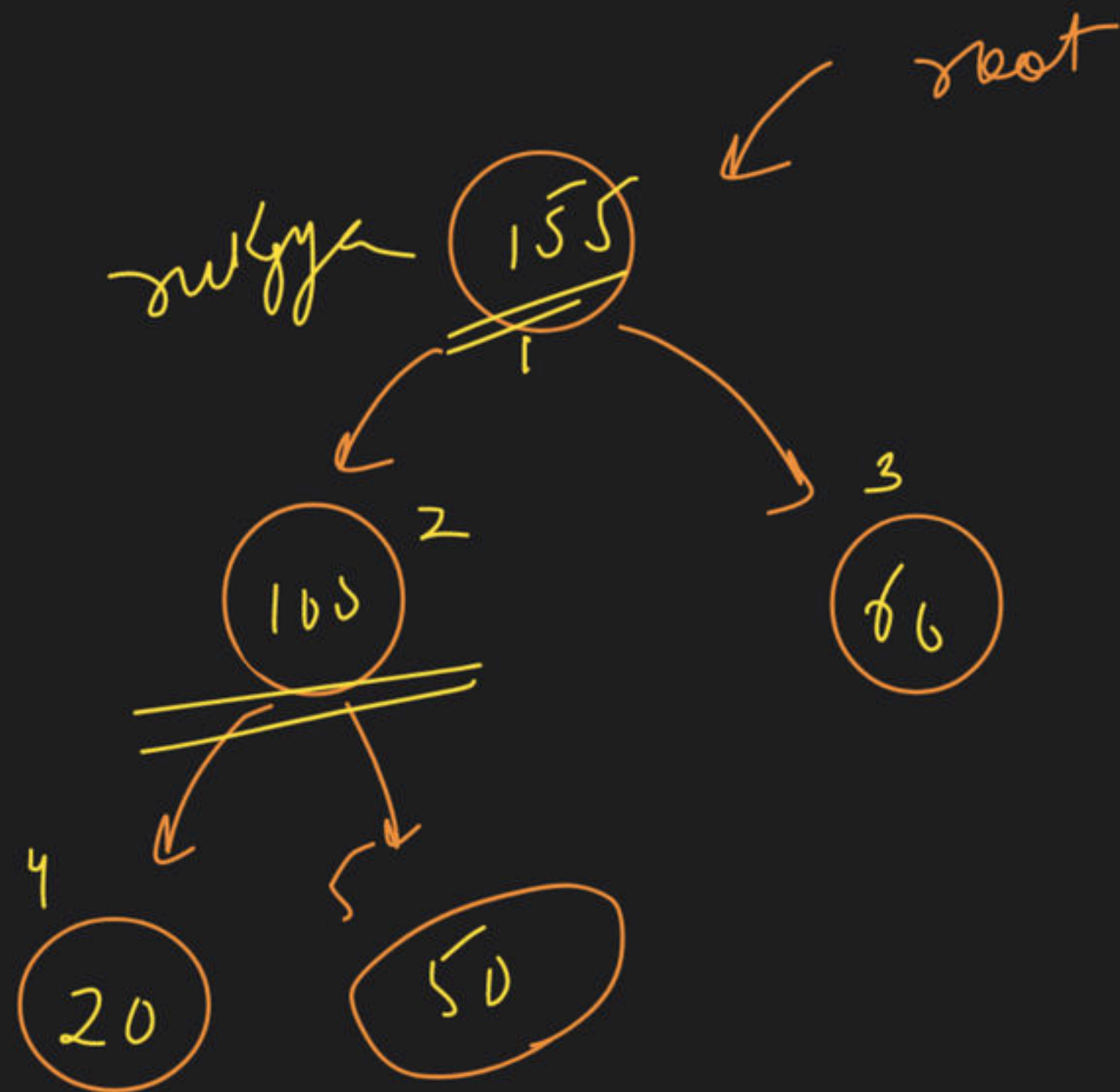
③ correct pos

I can {

- $\rightarrow p, \text{left}, \text{right}$
- $\rightarrow \text{cmp} \rightarrow \text{largest inden}$
- $\rightarrow \text{swap}$
- $\rightarrow \text{Recursion}$

insert(40)

$P > C$
 $i = 1$



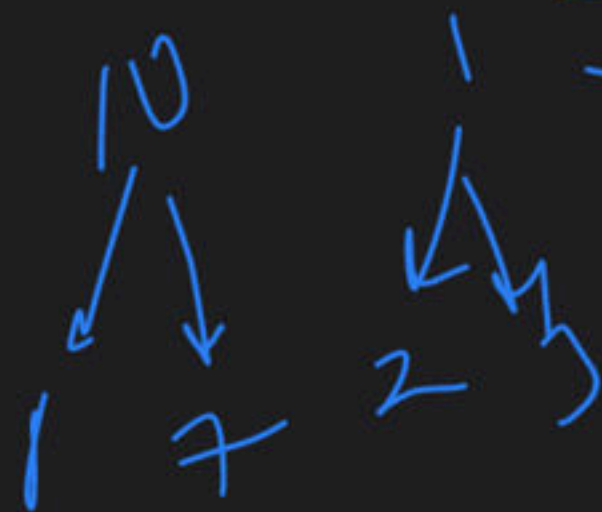
$\text{insert}(155)$

$$i = 5 \quad i/2 = 2$$

$$i = 2 \quad i/2 = 1$$

X	155	100	66	20	50
0	1	2	3	4	5

Heap → visualise

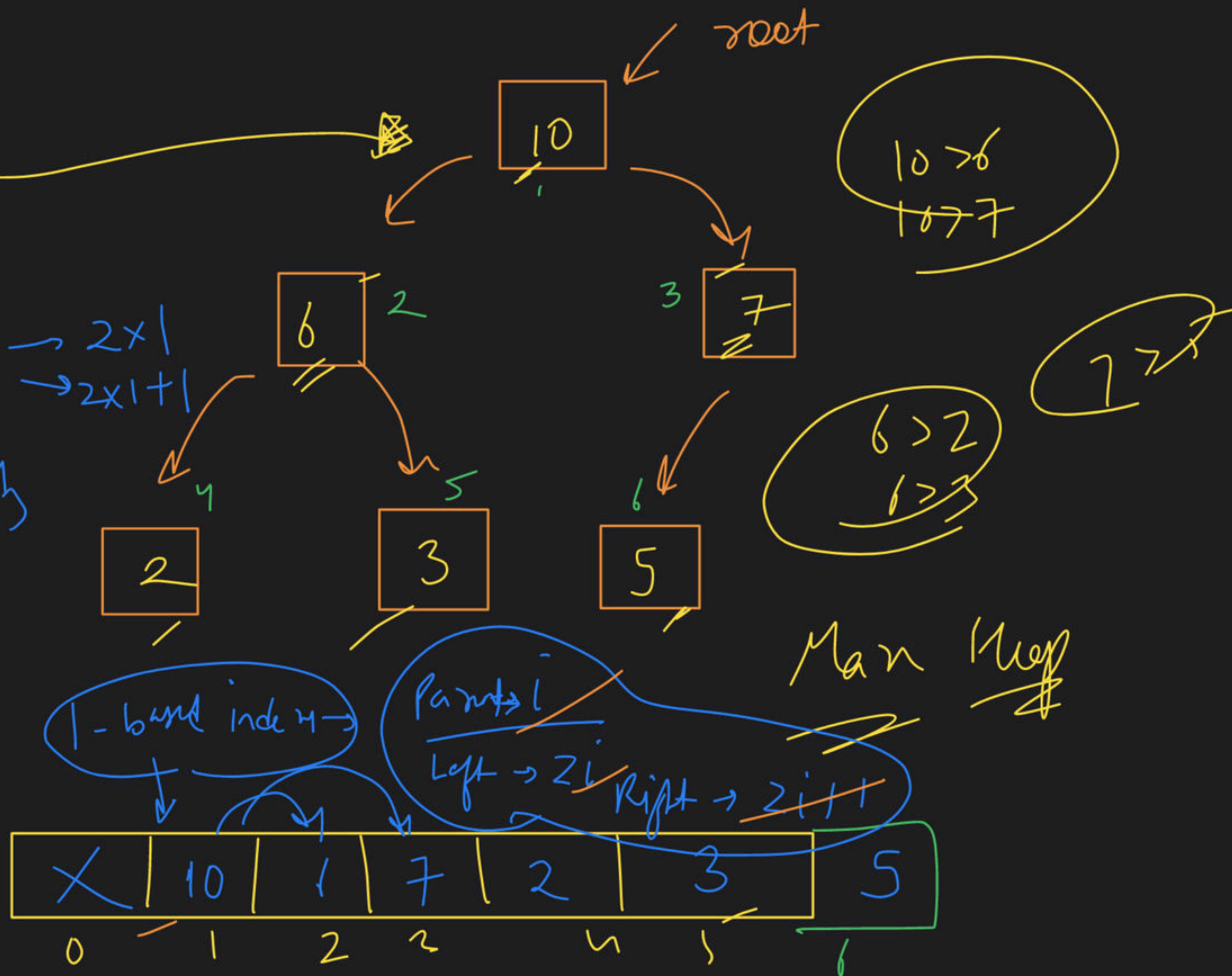


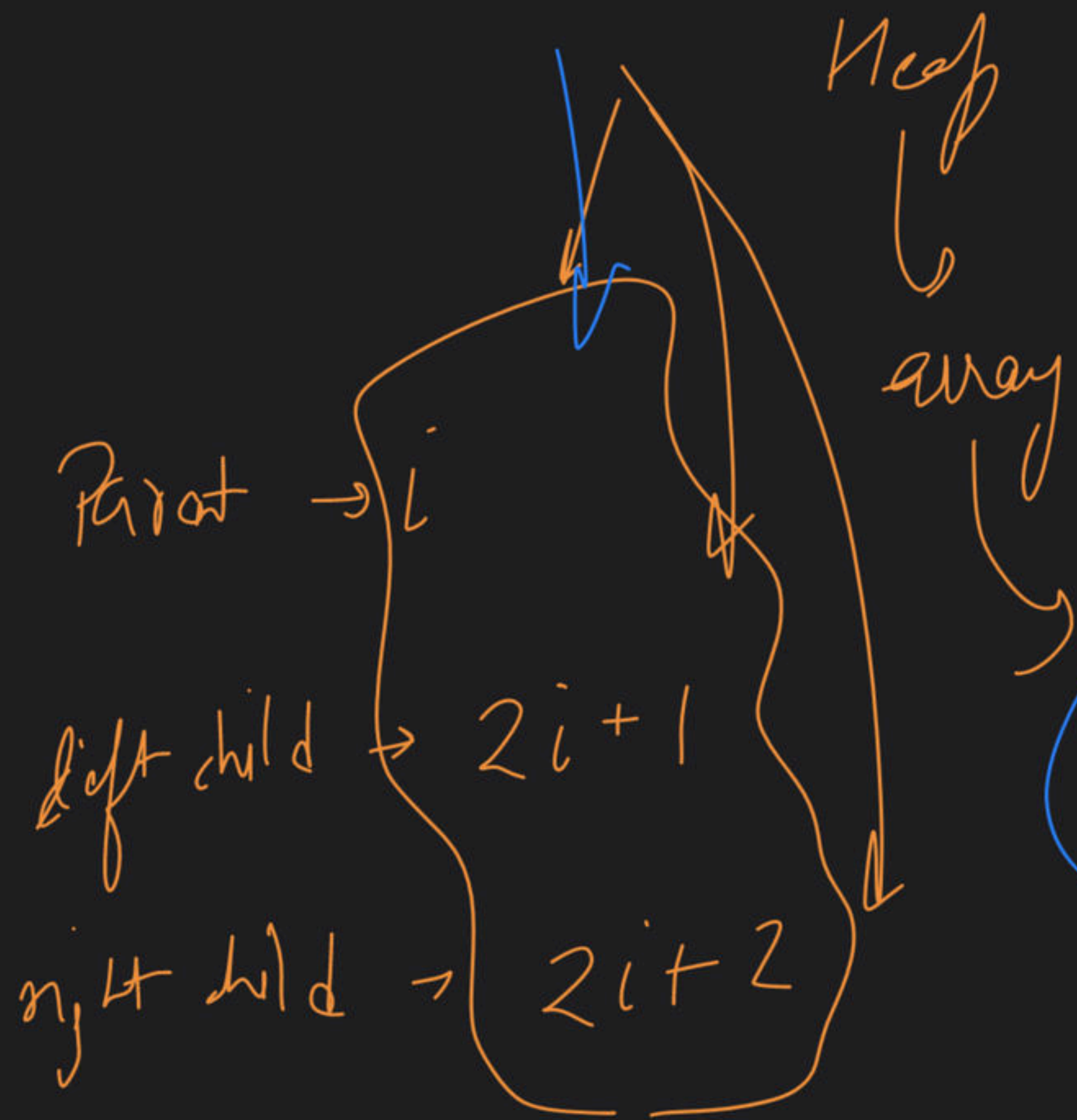
→ 2×1
→ $2 \times 1 + 1$

Implement

why

Array





0-indexing

